

The Islamia University of Bahawalpur



Software Design Document

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1 Introduction

The purpose of this software project is to develop an intelligent, user-friendly **AI-based Mental Health Monitoring Application** that assists users in self-monitoring their psychological well-being on a daily basis. With rising awareness and growing demand for accessible mental health tools, this app aims to bridge the gap between professional psychological care and personal mental wellness management.

This app is particularly designed for individuals who experience stress, anxiety, or mood fluctuations but may not have regular access to therapists or professional mental health services. It empowers users to become more aware of their emotional states through structured self-reporting and AI-powered analysis.

The app will collect daily mood ratings, short journal entries, and optional lifestyle information (like sleep hours or activity level) from users. It will then process this data using **Natural Language Processing (NLP)** and **Sentiment Analysis** techniques to provide feedback, emotional insights, and personalized mental health tips. The primary goal is to help users detect early warning signs of mental distress and promote mental resilience through regular reflection and actionable suggestions.

Additionally, the application ensures **data privacy, usability, and visual insight** by integrating secure storage (e.g., Firebase), a clean mobile interface (via Flutter), and user-friendly visualizations (charts and graphs) for trends in mood and behavior. Although not a substitute for medical diagnosis or treatment, the app serves as a valuable companion for users on their mental wellness journey.

The purpose of this document is to outline the complete functional and non-functional requirements of the AI-based Mental Health Monitoring Application, which is being developed as a final year project. This application is designed to promote mental well-being by helping users **track their emotional and psychological state** over time using a combination of self-reported data and AI-driven analysis.

In today's fast-paced and stressful world, mental health issues such as **stress, anxiety, depression, and burnout** are increasingly common. Unfortunately, due to societal stigma, lack of access to healthcare, or personal hesitation, many individuals do not seek professional help until their symptoms worsen. This app addresses this gap by offering a **non-judgmental, private, and easily accessible tool** for individuals to monitor and reflect on their mental health daily.

Key objectives of this system include:

- Enabling users to record their **daily mood, thoughts, stress levels, and habits** through quick, guided entries.
- Using **Artificial Intelligence**, particularly **Natural Language Processing (NLP)** and **Sentiment Analysis**, to evaluate user journal entries and detect underlying emotional patterns.
- Generating personalized mental health feedback and **self-help suggestions** based on user data.
- Visualizing user mood and stress trends over time with **interactive charts and analytics**.
- Encouraging consistent self-reflection to foster self-awareness, emotional regulation, and early identification of mental health concerns.

This app is not intended to replace professional psychological diagnosis or therapy but serves as a **complementary wellness tool**. It empowers users to take control of their emotional well-being by promoting healthy mental habits through regular check-ins, intelligent analysis, and supportive feedback.

The intended audience of this document includes developers, testers, academic supervisors, evaluators, and future maintainers of the project. The app is being developed using **Flutter** for cross-platform support (Android and iOS), **Firase** for backend services, and **Python-based AI models** for emotional analysis.

Scope:

The **AI-based Mental Health Monitoring App** aims to provide individuals with a comprehensive tool to monitor, track, and improve their mental well-being. By leveraging Artificial Intelligence (AI) and data analytics, the app helps users keep track of their emotional health through regular mood check-ins and journal entries. The app provides personalized feedback, mental health insights, and suggestions for coping strategies based on user input, ultimately empowering individuals to gain a deeper understanding of their mental state and improve their mental health over time.

The project focuses on **personalized mental health monitoring** by combining mood tracking, sentiment analysis of journal entries, and AI-driven feedback mechanisms. It is designed to help users be more aware of their emotional well-being and engage in self-care strategies in real-time, ensuring that they get the help they need before their mental health deteriorates.

Purpose of the Project:

The primary goal of the **AI-based Mental Health Monitoring App** is to offer an easy-to-use and reliable solution for individuals who want to improve their mental well-being. Mental health is often stigmatized, and many individuals struggle to assess or communicate how they feel. This app aims to overcome such barriers by providing a non-intrusive, safe platform to track moods, analyze emotional patterns, and receive expert-backed insights on improving mental health.

By encouraging daily self-reflection through journal entries and mood tracking, the app supports long-term mental health improvement through data-driven insights. The app will act as a personal mental health companion, providing users with personalized feedback, reminders for self-care practices, and even recommending resources like meditation exercises or mood-enhancing activities.

Project Scope:

The **scope** of this project includes the development and deployment of an **AI-based mobile application** (targeting iOS and Android platforms) designed to monitor and enhance mental well-being. The app will gather user data (mood, journal entries), analyze it using AI algorithms (sentiment analysis, trend detection), and provide **personalized feedback** and **mental health recommendations**.

The main features of the app include:

1. **Mood Tracking:**

- Users can select their current mood from a predefined set of moods (e.g., happy, sad, neutral, anxious, etc.) or use a sliding scale.

- Mood entries will be stored in a database, linked to timestamps for trend analysis.
- 2. **Journal Entry Analysis:**
 - Users can write journal entries about their day or experiences.
 - Each entry will undergo **sentiment analysis** to understand emotional tones (e.g., positive, negative, or neutral).
- 3. **AI-Powered Feedback:**
 - Based on mood and journal entries, AI algorithms will provide feedback and suggestions to help users manage their emotional state.
 - The system will offer personalized mental health resources, such as relaxation techniques, breathing exercises, or journaling prompts.
- 4. **Trend Analysis and Insights:**
 - Over time, the app will generate emotional **trend reports** showing users their mood patterns and emotional health trajectory.
 - The app will also suggest areas for improvement based on historical data (e.g., “You tend to feel more stressed on Mondays. Consider taking a mindfulness break on those days”).
- 5. **User Interaction and Data Privacy:**
 - Users can update their profile, add demographic information (e.g., age, gender), and set preferences for feedback and reminders.
 - Strong security and privacy measures will be implemented to protect sensitive user data (e.g., encrypted mood entries and journal text).

Scope in Detail:

In-Scope Features:

1. **User Registration and Authentication:**
 - Users must sign up with an email address, password, and optional demographic information.
 - User data will be stored securely and will allow personalized experiences.
2. **Daily Mood Logging:**
 - Daily mood tracking will be facilitated through a simple user interface where users can select their mood or write a quick note about how they feel.
3. **Sentiment Analysis of Journal Entries:**
 - The system will process journal text using sentiment analysis to detect emotional patterns.
 - Text analysis will include keyword detection for stress, happiness, sadness, and anxiety.
4. **Personalized Feedback and Recommendations:**
 - Based on mood entries and journal sentiment, users will receive AI-driven feedback.
 - Feedback could include mental wellness tips, coping strategies, and links to further resources (e.g., articles, videos, meditation exercises).
5. **Long-Term Trends and Insights:**
 - The app will generate visual reports and trends (e.g., line graphs, pie charts) showing users their emotional progress over time.
 - The app will suggest improvements, e.g., “You’ve been more anxious during the evenings. Try relaxation techniques before bed.”

6. Data Export and Backup:

- Users can export their mood and journal data in a readable format (e.g., PDF, CSV) for personal reference or sharing with a healthcare provider.
- Regular backups will ensure that user data is not lost.

Out-of-Scope Features:

1. Direct Mental Health Diagnosis:

- The app will not diagnose mental health conditions (e.g., depression, anxiety disorder). Instead, it will provide insights and guidance to help users reflect on their emotional health.

2. Real-time Therapy or Counseling:

- The app will not offer real-time therapy sessions. However, it will suggest resources such as online therapy, self-help tools, or contact information for healthcare providers.

3. Physical Health Monitoring:

- The app will focus on mental health only and will not track physical health metrics (e.g., step count, heart rate, etc.), except where relevant to emotional well-being (e.g., stress reduction exercises).

Limitations:

1. Internet Dependency:

- The app's AI-driven features (such as sentiment analysis and feedback generation) may require an internet connection for real-time processing.

2. Accuracy of AI Predictions:

- While the app uses advanced AI algorithms, the accuracy of mood predictions and feedback might be limited by the quality and quantity of user input.

3. User Commitment:

- The effectiveness of the app depends on how consistently users input their mood and journal data. Sporadic use may limit the app's ability to provide actionable insights.

Target Audience:

- **General Users:** Individuals interested in tracking and improving their mental health.
- **Mental Health Enthusiasts:** People who regularly practice self-care and wish to better understand their emotional well-being.
- **Individuals with Mild to Moderate Mental Health Concerns:** Users who may not yet seek professional help but want to monitor and improve their mental state.
- **Therapists/Healthcare Providers:** As the app allows for data export, it could be used by mental health professionals to track their patients' progress over time.

Conclusion:

The **AI-based Mental Health Monitoring App** is designed to bridge the gap between self-awareness and self-care, offering users a reliable tool to monitor their emotional health, gain insights, and adopt effective coping strategies. By integrating AI and sentiment analysis, this project aims to make mental health care more accessible and personalized for users of all backgrounds. Through continuous tracking, AI-driven feedback, and personalized suggestions, the app encourages users to take control of their mental well-being in a proactive, data-informed way.

2 Design methodology and software process model

2.1 Design Methodology

The design of this project follows a **user-centered and modular approach**, ensuring the system is intuitive, maintainable, and scalable. The application is divided into logical layers to separate concerns and allow parallel development.

a. Layered Architecture (High-Level Design)

1. **Presentation Layer (Frontend)**

- Developed using **Flutter**, it provides a responsive and consistent UI on both Android and iOS platforms.
- Features include user login, mood tracking interface, journal input, and visualization charts.

2. **Application Logic Layer**

- Manages the interaction between UI and backend services.
- Handles authentication, routing, AI request handling, and session management.

3. **AI & Analysis Layer**

- A Python-based micro service (or API) that receives journal entries and returns emotional analysis results (mood score, sentiment polarity, and mental wellness index).
- Built using NLP libraries like **NLTK**, **spacey**, or **Transformers** (e.g., BERT or Roberta).

4. **Data Storage Layer (Backend)**

- Uses **Firebase Real-time Database** and **Fire store** to store structured user data, mood logs, AI outputs, and feedback history.
- Firebase Authentication ensures secure user access and data privacy.

b. User-Centric Design (UX Approach)

The app is designed with the target user in mind — someone seeking emotional awareness, simplicity, and privacy. User feedback loops are embedded through:

- Minimalist UI/UX for ease of daily use
- Motivational messages
- Journaling prompts
- Analytics dashboards to empower user understanding

2.2 Software Process Model

To ensure systematic and iterative development, this project follows the **Iterative and Incremental Software Development Model**, specifically a simplified **Agile approach**, adapted for academic projects with limited resources and time.

Why Agile/Iterative?

- Frequent testing and feedback cycles
- Flexibility to incorporate improvements
- Phased delivery for early evaluations
- Rapid development in sprints

Development Phases

Phase	Description
Phase 1: Requirement Analysis	Gathering and finalizing functional and non-functional requirements. Identifying user needs and use cases.
Phase 2: Design	High-level architecture and UI wireframes. Selection of tech stack, AI model strategy, and database schema.
Phase 3: Implementation - Sprint 1	Core features: registration, login, mood input UI, Firebase integration.
Phase 4: Implementation - Sprint 2	AI analysis integration, journal logging, backend data flow, basic feedback generation.
Phase 5: Implementation - Sprint 3	Visualizations, notifications, performance enhancements, security features.
Phase 6: Testing & Debugging	Unit testing, UI testing, AI result validation, user testing.
Phase 7: Final Deployment	Publish app, prepare demonstration, finalize documentation.

Each sprint lasts 1–2 weeks and results in a working version with new features, tested and refined for the next cycle.

3 System overview

3.1 General Description

The AI-based Mental Health Monitoring App is a mobile-first wellness application designed to assist users in becoming more aware of their emotional and psychological state. It enables users to log their daily mood and thoughts, while an integrated AI engine analyzes this input to provide personalized mental health feedback. The core focus of the system is **early detection, self-reflection, and non-clinical mental wellness support**, particularly for individuals who may not have regular access to professional mental health services.

The system functions as a **daily mental check-in companion**, combining user interaction, emotion tracking, sentiment analysis, and data visualization — all delivered in a secure and private

mobile environment.

3.2 Functional Overview

The application provides the following major functionalities:

User Registration and Login: Secure sign-up and sign-in functionality using Firebase Authentication. Users can create an account and log in with their credentials.

1. **Mood Tracking:** Users select a mood level (e.g., from very happy to very sad) from a pre-defined scale or emoji-based interface. This is stored daily.
2. **Journal Entry:** Users can write a short reflection about their day. These entries serve as the input for AI analysis.
3. **AI Analysis of Journals:** An NLP engine processes journal entries using sentiment analysis and emotion detection techniques. The system assigns a mental health score based on the tone, emotional intensity, and recurring themes.
4. **Feedback and Suggestions:** Based on the AI output, the app provides personalized self-care suggestions, motivational quotes, mindfulness tips, and reflections to encourage mental well-being.
5. **Visualization of Trends:** The app displays the user's mood, AI scores, and journaling activity over time in the form of charts and graphs. These insights help users observe emotional patterns and triggers.
6. **Reminders and Notifications:** Gentle notifications remind users to check in daily without being intrusive.

3.3 System Context

This system is developed as a **cross-platform mobile application** using **Flutter**, which allows deployment on both Android and iOS devices. The backend infrastructure is powered by **Firebase**, which supports authentication, cloud data storage, and real-time syncing. The AI analysis component is implemented in **Python**, either as an integrated module or as an external API, depending on deployment requirements.

This project is developed in the context of:

- Growing demand for accessible and stigma-free mental health tools
- The academic setting of a final year AI project with a focus on **real-world AI application**
- The ethical requirement to prioritize **user privacy, empathy, and usability**

3.4 Design Summary

The system is designed using a **modular, layered architecture**, consisting of the following:

- **Frontend (Flutter UI):** User-facing screens for mood tracking, journaling, and data visualization.
- **Backend (Firebase):** Manages authentication, user profiles, mood logs, and AI results.
- **AI Engine (Python-based NLP):** Analyzes text input and generates emotional feedback.
- **Data Flow:** The app collects user input → stores it in Firebase → sends journal entries to the AI engine → receives mood scores and suggestions → updates user dashboards.

All user data is encrypted and stored securely, with a strong emphasis on ethical AI usage and user autonomy.

3.5 Background and Motivation

The idea for this system arises from the observation that while physical fitness apps are common, **mental health apps remain underutilized**, especially among youth in academic or high-stress environments. Many people experience mental health issues but hesitate to seek help due to cost, stigma, or lack of awareness.

This project aims to **empower users with AI**, not to diagnose, but to support and guide their emotional journey. By making mental health reflection easy, private, and intelligent, the system aspires to create a **positive daily habit** that improves emotional resilience over time.

4 Design models [along with descriptions]

4.1 Class Diagram (Object-Oriented Design)

The class diagram models the **static structure** of the system: classes, attributes, methods, and relationships between them (e.g., aggregation, composition, inheritance).

Key Classes in the System:

Class	Attributes	Methods	Relationships
User	user ID, name, email, password, mood History	login(), logout(), update Profile()	Composed of Mood and Journal
Mood Entry	moody, mood Level, timestamp, sentiment Score	addMood(), viewMood()	Associated with User
Journal Entry	entry ID, user Text, date, sentiment Result	writeEntry(), analyzeText()	Aggregates AIModel
AIModel	model Type, version	analyzeSentiment(text), returnFeedback()	Used by JournalEntry
Feedback	feedbackID, suggestionText, moodCategory	generateTip()	Depends on Sentiment Result

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Feedback	feedbackID, suggestionText, moodCategory	generateTip()	Depends on Sentiment Result

- User class **aggregates** Mood Entry and Journal Entry
- Journal Entry **uses (association)** AIModel to generate analysis
- AIModel is a **service class**, reusable
- Feedback is dynamically generated and **depends** on analysis results

✓ **Concepts used:** Aggregation (User–Journal), Association (Journal–AI), Composition (Mood Entry lifespan tied to User).

4.2 Sequence Diagram

This diagram shows the **interaction between objects over time**, useful for visualizing flow when a user submits a journal entry.

Use Case: Journal Entry + AI Analysis

Actors and Objects:

- User
- App UI
- Firebase
- AI API
- Feedback Generator

Sequence Steps:

1. User writes journal entry and submits.
2. App UI sends journal to Firebase (store).
3. App also sends journal text to AI API.
4. AI processes text → returns sentiment result and emotional tags.
5. App uses result to fetch appropriate suggestions.
6. Final analysis and suggestions are displayed to user.

✓ You should know how each message flows from user to AI and back.

4.3 State Transition Diagram

Useful for showing **different app states** depending on user actions.

Example: User Mental Check-in Flow

States:

1. Idle
2. Logged In
3. Mood Selected
4. Journal Written
5. AnalysisInProgress
6. Feedback Displayed
7. Session Ended

Transitions:

- On login → Idle → Logged In
- On mood input → Mood Selected
- On journal input → Journal Written
- On submit → AnalysisInProgress → Feedback Displayed → back to Logged In

✓ Understand **events** causing transitions and **final/initial** states.

4.4 Data Flow Diagram (DFD)

Level 0 (Context Diagram)

- **External Entities:** User
- **Process:** "Mental Health Monitoring System"
- **Data Stores:** User Data, Journal Entries, Mood History
- **Data Flows:** Mood Input, Journal Text, Feedback

Level 1: Key Processes

- 1.0 User Login/Register
- 2.0 Submit Mood
- 3.0 Write Journal
- 4.0 AI Analyzes Text
- 5.0 Generate Feedback
- 6.0 View Trends

Level 2 Example: Process 4.0 (AI Analysis)

- 4.1 Send Text to AI Engine
- 4.2 NLP Sentiment Analysis
- 4.3 Emotional Scoring
- 4.4 Store Result

✓ All DFD levels must show **input/output**, **processes**, **data stores**, and **entities** clearly.

5 Data design

The data design defines how information in the system is represented, stored, and processed. This app is **object-oriented** and uses a **cloud-based NoSQL database (Firebase Fire store)**, suitable for hierarchical, real-time, and scalable data operations on mobile devices.

5.1 Database Used:

- **Firebase Fire store (NoSQL, document-based)**
- **Firebase Authentication** (for secure login/registration)
- **Optional:** Local caching via **Hive** or **Shared Preferences** for offline support

5.2 Data Representation & Flow

Each **user** has a personal data space in Fire store, consisting of sub collections like:

- Mood Entries
- Journal Entries
- Feedback History

When a user performs a check-in:

1. **Mood and journal** are captured.
2. Journal text is sent to **AI API** for NLP processing.
3. Results (sentiment, emotion scores) are returned.
4. All data is stored under that user's profile in Fire store.

Entity	Description
User	Stores personal and login data; linked to all user-generated data.
Mood Entry	Captures the user's mood input (scale, emoji, notes).
Journal Entry	Stores the text of user reflection with AI-analyzed result.
AI Result	Contains output from the AI (mood score, sentiment label, emotional categories).

Entity	Description
Feedback	Stores generated tips or encouragement based on AI output.

5.1 Data Dictionary

Here is the alphabetically ordered data dictionary with **object-oriented structure** (object, attributes, methods):

1. AIModel

- **Attributes:**
 - `model Type (String)`: Type of NLP model used (e.g., BERT).
 - `version (String)`: Version of AI model.
- **Methods:**
 - `analyze Sentiment (text: String): Map<String, dynamic>`
 - `return Feedback (score: double): String`

2. Feedback

- **Attributes:**
 - `feedback ID (String)`: Unique feedback identifier.
 - `suggestion Text (String)`: AI-generated feedback message.
 - `mood Category (String)`: Mood type (e.g., Happy, Anxious, Neutral).
- **Methods:**
 - `generate Tip (score: double): String`

3. Journal Entry

- **Attributes:**
 - `entry ID (String)`: Unique journal entry ID.
 - `user Text (String)`: Written journal input by the user.
 - `date (Timestamp)`: Entry timestamp.
 - `sentiment Result (AI Result)`: Result from the AI analysis.
- **Methods:**
 - `write Entry (text: String): void`
 - `analyze Text (): AIResult`

4. Mood Entry

- **Attributes:**
 - `moody (String)`: Unique ID for each mood input.
 - `mood Level (Integer)`: Value from 1 (sad) to 5 (happy).
 - `timestamp (Timestamp)`: When mood was recorded.
 - `note (String)`: Optional notes or reasons.
- **Methods:**
 - `addMood (level: int): void`
 - `viewMoodHistory (): List<Mood Entry>`

5. User

- **Attributes:**
 - `user ID (String)`: Firebase-generated unique user ID.
 - `name (String)`: Full name of the user.
 - `email (String)`: User's email for login.
 - `mood History (List<Mood Entry>)`: All mood records.
 - `journal History (List<Journal Entry>)`: Journal entries with AI analysis.
- **Methods:**
 - `login(email, password): bool`
 - `logout(): void`
 - `update Profile(name): void`

Data design is a critical part of building your system, as it defines how data flows through your app, is stored, and is used for processing, analysis, and feedback generation. Here's a more detailed explanation of **how the data works** for your mental health app.

Data Transformation:

1. **User Data:**
 - **User Information:** When users sign up, their data will be collected (e.g., name, email, password, age, and other demographic information). This data will be stored in a **user database** for authentication and personalization purposes.
2. **Mood Data:**
 - Each user's **mood entries** (input through a slider or emoji's) will be stored along with a **timestamp** to track their emotional trends over time.
 - The mood data can be represented as numerical values (1-5) or categorical data (happy, sad, neutral, etc.).
 - **Mood Notes** will accompany mood entries, providing context or explanation for the mood selected.
3. **Journal Data:**

- **Journal Entries** are free-text data submitted by the user. These will be stored in a database as text.
- Sentiment analysis will process these journal entries to classify them based on sentiment (e.g., positive, neutral, or negative).
- The system will also track changes in writing patterns over time (e.g., positive language shift, usage of certain keywords).

4. **Feedback and Recommendations:**

- The **AI system** processes both mood and journal data to generate personalized feedback.
- Recommendations (e.g., mindfulness tips, relaxation exercises) will be dynamically generated based on user input and stored for later retrieval.

Data Storage:

1. **Database Structure:**

- **Relational Database** (e.g., MySQL, PostgreSQL) or **NoSQL Database** (e.g., MongoDB) could be used, depending on your choice of technology.
 - **User Table:**
 - Fields: `user_id`, `email`, `password`, `name`, `age`, `gender`, `registration_date`
 - **Mood Entry Table:**
 - Fields: `mood_id`, `user_id`, `mood_level` (1-5), `mood_note`, `timestamp`
 - **Journal Entry Table:**
 - Fields: `journal_id`, `user_id`, `journal_text`, `sentiment_score` (positive/neutral/negative), `timestamp`
 - **Feedback Table:**
 - Fields: `feedback_id`, `user_id`, `generated_feedback_text`, `timestamp`
 - **Trends Table:**
 - Fields: `trend_id`, `user_id`, `start_date`, `end_date`, `average_mood_score`, `average_sentiment_score`

2. **Data Flow:**

- Data is **submitted** by the user (e.g., mood, journal entries) through the app's interface.
- This data is **stored** in the database for later use.
- The **AI algorithm** retrieves the stored data to provide real-time **feedback** and generate **insights**.
- The **Trends** data is periodically updated based on user activity.

5.1 Data Dictionary

A **Data Dictionary** is a key tool in understanding the types and attributes of data used in your system. It provides a detailed list of all major system entities, along with their types and descriptions.

Here’s an example **Data Dictionary** for your app:

Entity Name	Attribute	Type	Description
User	user_id	Integer	Unique identifier for each user.
	email	String (Email)	User’s email address used for authentication.
	password	String (Encrypted)	User’s password (encrypted for security).
	name	String	User's full name.
	age	Integer	User’s age for age-based recommendations.
	gender	String (Optional)	User’s gender, used for personalized feedback.
	registration_date	Date Time	The date and time the user registered.
Mood Entry	mood_id	Integer	Unique identifier for each mood entry.
	user_id	Integer (FK)	Link to the user who made the mood entry.
	mood_level	Integer (1-5)	Mood level (1 = very sad, 5 = very happy).
	mood_note	String (Optional)	User's additional notes about their mood.
	timestamp	Date Time	The date and time when the mood entry was submitted.
Journal Entry	journal_id	Integer	Unique identifier for each journal entry.
	user_id	Integer (FK)	Link to the user who made the journal entry.
	journal_text	String	The text written by the user in their journal.
	sentiment_score	String (Positive/Neutral/Negative)	Sentiment score of the journal entry based on AI analysis.
	timestamp	Date Time	The date and time when the journal entry was submitted.
Feedback	feedback_id	Integer	Unique identifier for feedback entry.
	user_id	Integer (FK)	Link to the user who received the feedback.
	generated_feedback_text	String	The AI-generated feedback text based on mood and journal analysis.
	timestamp	Date Time	The date and time when the feedback was generated.
Trends	trend_id	Integer	Unique identifier for trends data.

Entity Name	Attribute	Type	Description
	user_id	Integer (FK)	Link to the user whose trends data is being tracked.
	start_date	Date Time	The start date for the trends data period.
	end_date	Date Time	The end date for the trends data period.
	average_mood_score	Float (1.0-5.0)	The average mood score for the user within the selected period.
	average_sentiment_score	Float (-1.0 to +1.0)	The average sentiment score for the user's journal entries within the period.

Data Processing and Organization:

1. Mood Data Processing:

- When the user submits their mood, it is saved in the **Mood Entry Table**.
- The app uses this data to calculate trends, average mood scores, and generate insights.

2. Sentiment Analysis of Journal Entries:

- AI Algorithms** will process the journal text (using natural language processing) to classify the sentiment as **positive**, **neutral**, or **negative**.
- Sentiment data is stored in the **Journal Entry Table** and used for **feedback generation** and **trends analysis**.

3. Trends Calculation:

- The app calculates **average mood scores** and **average sentiment scores** over a selected period (e.g., 7 days, 1 month) to display insights into the user's emotional progress.

4. Feedback Generation:

- Based on mood entries and journal sentiment, the app's AI engine generates personalized feedback (e.g., "You seem stressed today, consider practicing deep breathing exercises!").
- The **Feedback Table** stores these responses.

Example Data Flow:

1. User Interaction:

- User submits mood data (mood level and optional mood note).
- Mood data is saved in the **Mood Entry Table**.
- The system retrieves previous mood data to calculate trends.

2. Journal Entry Submission:

- User writes a journal entry.
- The system performs **sentiment analysis** on the journal text.
- The sentiment score is saved in the **Journal Entry Table**.

3. AI Feedback Generation:

- Based on mood and journal data, the app generates feedback.
- Feedback is saved in the **Feedback Table** and displayed to the user.

Next Steps:

- **Database Setup:** Choose a relational or NoSQL database, set up tables based on this design, and integrate it with your app's backend.
- **Data Validation:** Ensure proper data validation for mood levels, journal text, and sentiment analysis results.
- **User Data Security:** Implement **encryption** and **authentication mechanisms** for user privacy and security

6 Algorithm & Implementation

Each object and its key functions are explained using **pseudocode**, with **parameters**, **control flow**, **local data**, and logic broken down clearly.

1. User Object

login (email, password)

pseudocode

CopyEdit

Function login (email, password):

 Call FirebaseAuth.signInWithEmailAndPassword(email, password)

 If success:

 Return True

 Else:

 Show error message

 Return False

logout ()

pseudocode

CopyEdit

Function logout ():

 Call FirebaseAuth.signOut()

 Redirect to LoginScreen

update Profile(name)

pseudocode

CopyEdit

Function update Profile(name):

Get current user from FirebaseAuth
Update user profile name in Firestore

2. Mood Entry Object

addMood (level, note)

pseudocode

CopyEdit

```
Function addMood (level: Integer, note: String):  
    moody ← generateUUID ()  
    timestamp ← getCurrentTime ()  
    moodData ← {  
        "moody": moody,  
        "mood Level": level,  
        "note": note,  
        "timestamp": timestamp  
    }  
    Save moodData to Firestore under user's MoodEntries collection
```

viewMoodHistory ()

pseudocode

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```
Function viewMoodHistory ():  
    Fetch all documents from Firestore → MoodEntries collection  
    Return as List<Mood Entry>
```

3. Journal Entry Object

writeEntry(text)

pseudocode

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```
Function writeEntry (text: String):  
    entry ID ← generateUUID ()  
    date ← getCurrentDate ()  
    sentimentResult ← analyzeText(text)  
    journalData ← {  
        "entryID": entryID,  
        "userText": text,  
        "date": date,  
        "sentimentResult": sentimentResult  
    }  
    Save journalData to Firestore under JournalEntries
```

analyzeText (text)

pseudocode

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```
Function analyzeText (text: String):  
    Send HTTP POST request to AI API endpoint with {text: userText}
```

Receive JSON response with:
 sentiment Score, moodLabel, emotionCategories
Return result as AIResult object

4. AIModel Object

analyzeSentiment(text)

pseudocode

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```
Function analyzeSentiment (text: String):  
    Clean and preprocess input text  
    Tokenize and vectorize text  
    Predict sentiment using pretrained NLP model  
Return:  
    sentiment Score: Float  
    moodLabel: String (e.g., Happy, Sad, Angry)  
    emotions: List<String> (e.g., Anxiety, Calm, etc.)
```

return Feedback(score)

pseudocode

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```
Function returnFeedback (score: Float):  
    If score > 0.7:  
        Return "You're doing well! Keep it up."  
    Else if score between 0.4 to 0.7:  
        Return "You're doing okay. Try relaxing activities."  
    Else:  
        Return "You may be feeling low. Take a walk or talk to  
someone you trust."
```

5. Feedback Object

generate Tip(score)

pseudocode

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```
Function generateTip (score: Float):  
    Use score to determine user mood range  
    Select tip from pre-defined list based on moodCategory  
Return suggestionText
```

Additional Notes

- All **UUIDs** are used to uniquely identify mood and journal entries.
- All **timestamps** are generated using system time in UTC.
- AI analysis is done using **REST API** or **local inference** if bundled model is used.
- User data is **encrypted** and stored using Firebase Security Rules.

7 Software requirements traceability matrix

Table 7: Requirements Traceability Matrix

Req.Number	Ref. Item	Design Component	Component Items
FR01	Mood Tracking	Class Diagram	Mood Entry class, addMood(), viewMoodHistory()
FR02	User Login	Class Diagram	User class, login(), logout()
FR03	Journal Writing	Class Diagram	JournalEntry class, writeEntry(), analyzeText()
FR04	AI Sentiment Analysis	AIModel Class	analyzeSentiment(), returnFeedback()
FR05	Data Persistence	Data Flow Diagram (DFD)	Level 1: Processes 2.0-5.0, Firebase Integration
FR06	Personalized Feedback	Feedback Class	generateTip()
FR07	Visualization of Trends	Class Diagram	

	Ref. Item	DFD Diagram Number/Level	FunctionName(s)
Req. Number			
FR01	Mood Tracking	DFD Level 1	addMood(), viewMoodHistory()
FR02	User Login	DFD Level 1	login(), logout()
FR03	Journal Writing	DFD Level 1	writeEntry(), analyzeText()
FR04	AI Sentiment Analysis	DFD Level 1	analyzeSentiment(), returnFeedback()
FR05	Data Persistence	DFD Level 2	Firebase storage, Cloud functions
FR06	Personalized Feedback	DFD Level 1	generateTip()
FR07	Visualization of Trends	DFD Level 1	Visualization logic, Firebase charts

Explanation:

- **Req. Number:** Refers to the requirement number as listed in your Functional Requirements section.
- **Ref. Item:** Identifies the specific feature or component that is being referenced.
- **Design Component:** This column links the requirement to the specific design components (e.g., class diagram, DFD, etc.).
- **Component Items:** Identifies the relevant classes, methods, or functions that fulfill the requirement.

8 Human interface design

This section describes how users will interact with your mental health monitoring app and what the **user interface (UI)** will look like. The goal is to ensure that users can easily access features such as mood tracking, journaling, sentiment analysis, and personalized feedback.

8.1 Screen Images

General Overview of App Screens:

1. **Welcome Screen:**
 - A clean, welcoming screen with the app logo and options for **Login** or **Sign Up**.
2. **Login/Sign Up Screen:**
 - Inputs: **Email, Password** (both fields with validation)
 - Action Buttons: **Login** (user authentication) or **Sign Up** (new account creation).
3. **Home Screen:**
 - After login, the user is greeted with their **dashboard**.
 - **Buttons for accessing** mood input, journal writing, and viewing feedback.
4. **Mood Entry Screen:**
 - The user is asked to input their mood using a **slider** or **emoji scale** (1-5).
 - Optional: The user can add a **mood note** to provide context for the mood selection.
5. **Journal Writing Screen:**
 - The user can type reflections or thoughts into a **text box**.
 - **Submit button** triggers AI analysis of the journal text.
6. **Feedback Screen:**
 - After submitting the journal, the app displays **AI-generated feedback** based on mood and sentiment analysis.
 - **Suggestions for improvement**, such as relaxation techniques or social support tips.
7. **Trends & Insights Screen:**
 - **Graphical representation** of the user's mood over time.
 - **Charts/Graphs** for users to visualize their emotional journey.

Note:

I can generate visual **screen mockups** based on these descriptions, using any design tool or diagrams.

8.2 Screen Objects and Actions

Below is a discussion of the **screen objects** (UI components) and the **actions** associated with each:

1. Welcome Screen:

- **Objects:** Logo, Login Button, Sign Up Button
- **Actions:**
 - Clicking **Login** takes the user to the **Login Screen**.

- Clicking **Sign Up** takes the user to the **Sign Up Screen**.

2. Login Screen:

- **Objects:** Email Field, Password Field, Login Button, Sign Up Link
- **Actions:**
 - **Email & Password fields:** Users input login credentials.
 - **Login Button:** Submits data for authentication. If successful, users proceed to the **Home Screen**.
 - **Sign Up Link:** Navigates to **Sign Up** screen.

3. Home Screen:

- **Objects:** Mood Input Button, Journal Button, View Feedback Button, Trend Graph Button
- **Actions:**
 - Clicking **Mood Input** takes the user to the **Mood Entry Screen**.
 - Clicking **Journal Button** navigates to the **Journal Writing Screen**.
 - Clicking **View Feedback** leads to the **Feedback Screen**.
 - Clicking **Trend Graph** shows the **Trends & Insights** screen.

4. Mood Entry Screen:

- **Objects:** Mood Slider (1–5 scale), Mood Emoji Picker, Mood Note Text Box, Submit Button
- **Actions:**
 - **Mood Slider/Emoji Picker:** Users select their mood level (1-5 scale or emojis).
 - **Mood Note:** User can optionally add a description or context.
 - **Submit Button:** Saves mood data and triggers the display of feedback or journaling options.

5. Journal Writing Screen:

- **Objects:** Journal Text Box, Submit Button
- **Actions:**
 - **Text Box:** User writes reflections or a journal entry.
 - **Submit Button:** Submits text for **AI sentiment analysis** and saves it to the database.

6. Feedback Screen:

- **Objects:** AI-Generated Feedback Text, Suggestions List, Return to Home Button
- **Actions:**
 - The **feedback text** displays the results of sentiment analysis (e.g., “You are feeling anxious. Consider relaxation exercises.”).
 - **Suggestions List** provides additional guidance based on mood (e.g., mindfulness tips, journaling ideas).
 - **Return to Home** button brings users back to the **Home Screen**.

7. Trends & Insights Screen:

- **Objects:** Mood Graph, Date Range Selector, Insights Text Box
- **Actions:**
 - The **Mood Graph** displays a chart of mood changes over time.
 - The **Date Range Selector** allows users to choose which time period to view.
 - **Insights Text Box** displays a summary of emotional trends (e.g., “You had more positive moods in the last month. Keep up the good work!”).

Appendix I: UML Design, ER Diagram, and Data Flow Diagrams

For further details on UML (Object-Oriented Design), ER diagrams, and data flow, please follow the linked references in your document to study the **UML guidelines** (<http://agilemodeling.com/artifacts/>) and **Data Flow Diagrams** (<http://www.agilemodeling.com/artifacts/dataFlowDiagram.htm>).

In this section, we focus on the **user interface (UI)** of the AI-based Mental Health Monitoring App, aiming to ensure a smooth, user-friendly experience. Each screen is designed with the user’s well-being in mind, prioritizing ease of use and accessibility.

8.1 Screen Images

Let’s break down the **screens** and **functions** that will be shown to the user. The design of each screen focuses on simplicity, clarity, and ease of navigation:

1. Welcome Screen:

- **Purpose:** Greets the user and provides entry points to login or sign up.
- **Components:**
 - **App Logo:** Positioned at the top of the screen.
 - **Login Button:** Takes the user to the login screen for authenticated users.
 - **Sign Up Button:** Directs to a registration page for new users.
- **Actions:**
 - Clicking **Login** navigates to the **Login Screen**.
 - Clicking **Sign Up** navigates to the **Sign-Up Screen**.

2. Login/Sign-Up Screen:

- **Purpose:** Allows users to authenticate themselves or create a new account.
- **Components:**
 - **Email Field:** Input field for the user’s email.
 - **Password Field:** Input field for the user’s password.
 - **Login Button:** Submits login credentials for authentication.
 - **Sign Up Link:** For new users to register an account.
- **Actions:**
 - Clicking **Login** validates credentials and proceeds to the **Home Screen** if successful.

- Clicking **Sign Up** leads to the registration form where the user enters additional personal information.

3. Home Screen:

- **Purpose:** The dashboard for all activities.
- **Components:**
 - **Mood Tracking Button:** Navigates to the **Mood Entry Screen**.
 - **Journal Writing Button:** Navigates to the **Journal Writing Screen**.
 - **View Feedback Button:** Takes the user to the **Feedback Screen** for mood insights.
 - **View Trends Button:** Shows the **Trends & Insights** screen for mood history.
- **Actions:**
 - Clicking the **Mood Tracking Button** directs the user to record their mood.
 - Clicking the **Journal Writing Button** brings the user to a place to write journal entries.
 - Clicking the **View Feedback Button** displays AI-driven personalized feedback.
 - Clicking the **View Trends Button** shows historical mood data visualized as graphs or charts.

4. Mood Entry Screen:

- **Purpose:** Enables users to track their emotional state.
- **Components:**
 - **Mood Slider:** A 1–5 scale slider or emoji set to express mood (e.g., happy, neutral, sad).
 - **Mood Note:** An optional text box where the user can provide additional context.
 - **Submit Button:** Saves the mood entry and proceeds to the **Feedback Screen**.
- **Actions:**
 - The **Mood Slider** or emoji set allows the user to visually express their mood.
 - The **Mood Note** field is optional but encourages the user to provide context for their mood.
 - Clicking **Submit** saves the mood entry to the database and triggers a feedback response from the AI.

5. Journal Writing Screen:

- **Purpose:** Allows the user to write daily reflections and thoughts.
- **Components:**
 - **Text Box:** A large input field for free-text journaling.
 - **Submit Button:** Sends the journal text for AI sentiment analysis and saves it in the database.
- **Actions:**
 - The **Text Box** allows users to freely express thoughts or emotions.
 - Clicking **Submit** invokes AI analysis of the text and provides personalized feedback based on the content.

6. Feedback Screen:

- **Purpose:** Displays AI-generated feedback based on the user's mood and journal entries.
- **Components:**
 - **Feedback Text:** The response generated by the AI (e.g., "You seem anxious, try a mindfulness exercise").
 - **Suggestions:** A list of tips or recommendations tailored to the user's mood (e.g., deep breathing exercises, talking to a friend).

- **Home Button:** Allows the user to return to the home screen.
 - **Actions:**
 - The feedback text is updated dynamically based on the user's entries.
 - **Suggestions** provide helpful resources or coping strategies based on sentiment analysis results.
 - Clicking **Home** returns the user to the **Home Screen** for further actions.
7. **Trends & Insights Screen:**
- **Purpose:** Visualizes the user's emotional journey over time.
 - **Components:**
 - **Mood Graph:** A chart showing mood ratings over a selected time range (e.g., past 7 days, past month).
 - **Date Range Selector:** A dropdown or slider to filter the data (e.g., by week, month).
 - **Insights Text:** Summary of trends based on the mood data (e.g., "Your mood was significantly better this week. Great progress!").
 - **Actions:**
 - The **Mood Graph** updates based on the selected time range.
 - **Insights Text** offers an overview of the user's mood trends, acknowledging improvements or highlighting patterns.
 - The user can interact with the graph to view specific data points.

8.2 Screen Objects and Actions

This section discusses in detail the **screen objects** (UI elements) and their **associated actions**, outlining how each component behaves when the user interacts with it:

1. Welcome Screen:

- **Objects:**
 - **Logo:** Displays the app's brand.
 - **Login Button:** Starts the authentication process.
 - **Sign Up Button:** Leads to the account creation process.
- **Actions:**
 - The **Login Button** opens the **Login Screen** where the user can input their credentials.
 - The **Sign Up Button** opens the **Sign Up Screen** for new users.

2. Login Screen:

- **Objects:**
 - **Email Input:** The user types their email to authenticate.
 - **Password Input:** The user types their password.
 - **Login Button:** Submits the login credentials to authenticate.
 - **Sign Up Link:** Navigates to the **Sign Up Screen**.
- **Actions:**
 - **Login Button** verifies credentials against the database and transitions to the **Home Screen**.

- If the credentials are incorrect, an error message is displayed, and the user is asked to try again.
- **Sign Up Link** navigates to a screen where the user creates a new account.

3. Home Screen:

- **Objects:**
 - **Mood Entry Button:** Takes the user to the **Mood Entry Screen**.
 - **Journal Button:** Opens the **Journal Writing Screen**.
 - **View Feedback Button:** Shows AI-generated feedback based on user inputs.
 - **Trends Button:** Displays the **Trends & Insights Screen**.
- **Actions:**
 - Clicking any of these buttons takes the user to the corresponding screen to input data or view insights.

4. Mood Entry Screen:

- **Objects:**
 - **Mood Slider:** User selects their mood on a scale of 1-5.
 - **Mood Emoji:** Users choose an emoji that best represents their mood.
 - **Mood Note:** Users can add any context or additional thoughts about their mood.
 - **Submit Button:** Saves the mood data.
- **Actions:**
 - The mood is stored in the database when the **Submit Button** is clicked.
 - A confirmation message is shown once the mood is successfully saved.

5. Journal Writing Screen:

- **Objects:**
 - **Text Box:** Where users can type their daily journal entry.
 - **Submit Button:** Submits the journal entry for AI analysis.
- **Actions:**
 - When the user presses **Submit**, the app sends the journal text to the AI for analysis and then stores the entry.

6. Feedback Screen:

- **Objects:**
 - **Feedback Text:** Displays AI's feedback based on the user's input.
 - **Suggestions List:** Lists recommended actions based on the analysis.
- **Actions:**
 - The feedback is dynamically updated based on the user's entries.
 - **Suggestions** provide helpful insights or self-care tips for the user.

7. Trends & Insights Screen:

- **Objects:**
 - **Mood Graph:** A line chart showing mood progression over time.
 - **Date Range Selector:** A dropdown or slider to filter the data.
 - **Insights Text:** A brief summary of the user's emotional trends.
- **Actions:**
 - The **Mood Graph** updates in real-time as the user selects different date ranges.